PROJECT UDYAM DIGITAL: DATA PRIVACY & SECURITY NOTE

Technical Architecture Alignment with India's Digital Personal Data Protection (DPDP) Act, 2023

Track: Airtel SYNC 2026 – Digital Track | **Entity:** MSME Acquisition Engine | **Date:** November 2026 | **Status:** Technical Architecture Specification

1. EXECUTIVE SUMMARY & STATUTORY SCOPE

Project Udyam Digital implements concrete **Privacy-by-Design** software controls guided by the principles of India's *Digital Personal Data Protection (DPDP) Act, 2023*. The platform enables frictionless onboarding for Indian MSMEs while safeguarding proprietor personal data, statutory identity artifacts, and utility documents. ***Engineering Disclosure:*** *This document outlines concrete software controls implemented in the prototype; it constitutes an architectural blueprint and proof-of-concept implementation, not formal statutory legal certification. All registry lookups query indexed synthetic demonstration benchmarks.*

2. STATUTORY PRINCIPLE TO TECHNICAL CONTROL MAPPING

DPDP Act Principle	Statutory Intent	Implemented Technical Architecture Safeguard
Notice & Affirmative Consent (Section 6)	Clear, unbundled, multilingual notice prior to personal data collection.	Explicit, unbundled opt-in checkboxes presented in proprietor's selected language (Hindi/English). Immutable consent logging with ISO 8601 UTC timestamp and one-way SHA-256 client IP hash.
Purpose Limitation (Section 5)	Data processed strictly for specified, lawful provisioning purpose.	KYC documents and registry data are processed strictly for statutory telecom KYC and premises feasibility. Data is strictly cordoned off from secondary advertising or commercial telemetry.
Data Minimisation (Section 6(1))	Collect solely the minimum necessary personal data.	No GST certificate is requested for retail broadband SKUs (where requires_gstin=='N'). Only minimum statutory identifier and supply address are captured for service delivery.
Storage Limitation (Section 8(7))	Retain personal data only as long as required for purpose.	Documents are processed in memory by the hosted backend and are not forwarded to external third-party AI APIs. Production deployment requires further security and retention assessment.
Reasonable Security (Section 8(5))	Protect against data leaks, breach, and unauthorized access.	No customer documents are forwarded to external third-party AI APIs; in-memory computer-vision image-quality pre-checks (Laplacian blur QA). Mandatory PII masking (Aadhaar XXXX-XXXX-1234, phone XXXXXX9821).

3. CORE ARCHITECTURAL PRIVACY BOUNDARIES

A. No External Third-Party AI API Forwarding: Documents are processed in memory by the hosted backend and are not forwarded to external third-party AI APIs. Production deployment requires further security and retention assessment. The conversational AI Advisor is a deterministic, rule-grounded needs mapper strictly isolated from document processing, while image-quality checks, aspect analysis, and blur detection execute in memory on the backend.

B. Ephemeral Document Processing & Roadmap Controls: Customer utility bills and identity documents are processed in-memory during session validation. For commercial production scale, the architecture specifies HSM envelope encryption (AES-256-GCM) with automated cryptographic shredding at T+30 days post-activation, retaining only statutory Department of Telecommunications (DoT) compliance hashes.

C. Comprehensive PII Masking & Log Sanitation: Customer phone numbers, Signatory Aadhaar references, and tax numbers are dynamically masked across client-facing screens, partner/field consoles, and server logs. Middleware sanitizes all JSON output to prevent accidental exposure of raw personal data in application telemetry.

D. Granular Role-Based Access Control (RBAC): Proprietors access order status strictly via authenticated session tokens. Assisted digital channel partners and field executives view only aggregated drop-recovery queues and progress milestones, completely barred from viewing underlying customer identification files.